

Computational Social Science with Images and Audio

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Question and Approach

- Economics has formal procedures for **testing** hypotheses.
- The process that produces those hypotheses remains largely informal: observation, intuition, and creativity.
- High-dimensional data now contain patterns that people may never notice directly.
- The paper combines a predictor, a generative model, and human interpretation to turn opaque signal into candidate hypotheses.

Can a black-box predictor help people generate novel, interpretable hypotheses?

What the Paper Does

1. **Setting:** Link mugshots to pre-trial detention decisions for 51,751 arrests in Mecklenburg County, North Carolina.
2. **Prediction:** Train an image model to measure how much judges' decisions can be predicted from facial appearance.
3. **Visualization:** Use a generative model to create realistic face pairs that differ in predicted detention risk.
4. **Interpretation and testing:** Ask people to name the visual differences, then measure those features in real mugshots and test whether they predict actual decisions.

Mugshots → image prediction → synthetic contrasts → named hypotheses

Main Findings

1. Mugshots alone capture **27% to 45%** of the predictable variation in detention decisions, depending on the accuracy metric.
2. Measured demographics, perceived traits, and human predictions explain only a small share of the image model's predictive signal.
3. Subjects learn to interpret detention-risk morphs and name two features:
 - **well-groomed**
 - **heavy-faced** or full-faced
4. Both features predict actual judge decisions and survive several alternative explanations.

The output is a hypothesis to test, not a causal conclusion.

Outline

- 1 A Framework for Discovery
- 2 Application and Data
- 3 A Surprising Source of Predictive Signal
- 4 Algorithm-Human Communication
- 5 Evaluating the Generated Hypotheses

Hypothesis Generation and Hypothesis Testing

Generation

- Produces a candidate relationship $h(x)$
- Searches a large space of possible ideas
- Uses patterns in observed data
- Values novelty and interpretability

Testing

- Evaluates a prespecified $h(x)$
- Requires a separate identification strategy
- Distinguishes correlation from causation
- Values size, precision, and robustness

The paper improves the first stage without weakening the standards of the second.

Criteria for a Discovery Procedure

Let P be a discovery procedure. Given $D \sim \mathcal{D}$, it returns $h = P(D)$: a candidate interpretable function that maps x into a prediction of y .

- **Comprehensibility:** $\xi(P) = \Pr(h \in \mathcal{I})$
 - $\mathcal{I}$ is the set of hypotheses people can understand and describe.
- **Plausibility:** $\pi(P) = \Pr(\text{cor}(h(x), y) > 0)$
 - $h(x)$ is the candidate prediction for input x , and y is the observed outcome. Their correlation must have the predicted sign.
- **Replicability:** $\rho(P) = \Pr(P(D) = P(D'))$
 - Independent draws D and D' produce the same hypothesis.

Probabilities are over repeated draws from $\mathcal{D}$. Causality requires a separate test.

Formal Setup

Suppose $x \in \mathbb{R}^k$, $y \in \{0, 1\}$, and the data follow $y = f^*(x)$.

Application: x is a mugshot represented by pixels, and $y = 1$ means the judge detains the defendant.

- $\mathcal{H}$: all possible functions mapping x into y
- $\mathcal{I} \subseteq \mathcal{H}$: functions with a human-understandable description
- $\Phi^* = \{\phi_1^*, \dots, \phi_j^*\} \subset \mathcal{I}$: interpretable components of the truth

$$f^*(x) = \underbrace{\phi_1^*(x) + \dots + \phi_j^*(x)}_{\text{interpretable component}} + \Delta^*(x).$$

The discovery problem searches for $\Phi \subset \mathcal{I}$ such that

$$f_\Phi(x) = \sum_{\phi \in \Phi} \phi(x)$$

approximates f^* well enough to justify further testing.

The Morphing Discovery Procedure

Let $p(x)$ describe the support of realistic inputs and draw an initial point $x_0 \sim p$.

$m(x)$: model-predicted detention probability $\underline{m}$: low cutoff $\overline{m}$: high cutoff

$$x^- = \arg \min_x \{ \|x - x_0\| : m(x) \leq \underline{m}, p(x) > 0 \},$$

$$x^+ = \arg \min_x \{ \|x - x_0\| : m(x) \geq \overline{m}, p(x) > 0 \}.$$

Application: x^- is the nearest realistic mugshot with low predicted detention, and x^+ the nearest with high predicted detention.

1. Construct pairs (x^-, x^+) that remain realistic and differ strongly in $m(x)$.
2. Show many pairs to independent human subjects.
3. Ask subjects to name an interpretable feature $h \in \mathcal{I}$ that distinguishes the pair.
4. Verify that $h(x)$ predicts both $m(x)$ and the original outcome y .

What the Formal Result Establishes

Under restrictions on the number of pairs people inspect, problem complexity, and the frequency with which false hypotheses fit the morph pairs:

$$\pi(P_m) \geq \pi(P_c), \quad \rho(P_m) \geq \rho(P_c), \quad \xi(P_m) = \xi(P_c) = 1.$$

- P_m : morphing-based discovery procedure
- P_c : human creativity applied directly to raw observations
- Nearest counterfactuals reduce irrelevant differences.
- Out-of-sample prediction reduces the chance that people name sample-specific noise.

Assumptions

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Setting and Sample

- Within 24–48 hours of arrest, a judge decides whether the defendant waits for trial in jail or at home.
- The authors link Mecklenburg County arrests, judge decisions, prior records, and frontal mugshots.
- 81,166 arrests yield **51,751 usable cases** involving 42,353 unique defendants.
- Arrestee-level splits prevent leakage across repeated arrests:

Sample	Cases	Share
Training	23,138	44.7%
Validation	9,604	18.6%
Lock-box	19,009	36.7%

The lock-box is large because it includes the final six months plus 30% of earlier arrestees.

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Prediction and Measurement

For each case i , let $y_i = 1$ if the judge detains the defendant.

Structured model

$$m_s(x_i) = \widehat{\text{Pr}}(y_i = 1 \mid X_i^s)$$

X_i^s : charge, record, age, and gender

Gradient-boosted decision trees

The stacked predictor combines both signals:

$$m_p(x_i) = \widehat{\beta}_s m_s(x_i) + \widehat{\beta}_u m_u(x_i).$$

Independent workers label demographics, perceived psychological traits, and incentivized guesses about which matched defendant the judge detained.

Unstructured model

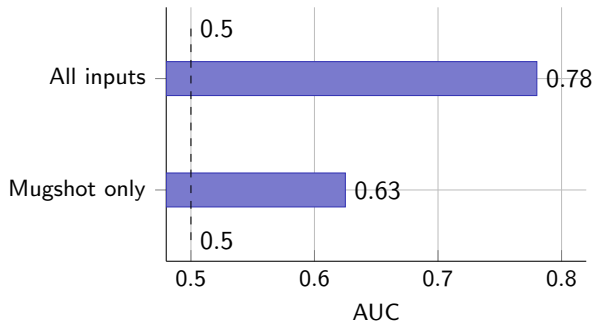
$$m_u(x_i) = \widehat{\text{Pr}}(y_i = 1 \mid X_i^u)$$

X_i^u : mugshot pixels

Convolutional neural network

[Model Details](#)

The Face Contains a Large Share of Predictable Signal



AUC decomposition

$$\frac{0.625 - 0.500}{0.780 - 0.500} = 44.6\%.$$

R^2 decomposition

$$\frac{0.03}{0.11} = 27.3\%.$$

The bottom-to-top mugshot-prediction quartile gap is **20.4 pp**, compared with 4.8 pp for violent versus non-violent charges.

Known Features Explain Little of the Algorithm's Signal

Predictors	Outcome	R^2
Known facial features together	Algorithm prediction	0.2228
Algorithm prediction alone	Judge decision	0.0331
Known facial features together	Judge decision	0.0162
Human guesses alone	Judge decision	0.0025

- The coefficient on $m_u(x)$ falls only from 0.6963 to 0.6171 after all known controls enter.
- Human subjects predict the detained image correctly only 51.4% of the time.

Regression Table

Does the Face Proxy for Re-arrest Risk?

If appearance affects detention only because it predicts re-arrest, then the risk gradient required to rationalize judge behavior is

$$\frac{\partial \Pr(\text{detain}) / \partial m_u}{\partial \Pr(\text{detain}) / \partial \widehat{\text{risk}}} = \frac{0.6963}{0.6103} = 1.141.$$

Among released defendants, the observed mugshot-risk gradient is only 0.6172.

To close the gap, the unobserved gradient among detained defendants would need to equal

$$0.767(0.6172) + 0.233X = 1.141 \quad \Rightarrow \quad X = 2.985.$$

The paper treats this as suggestive, not definitive: re-arrest is itself selected and measured through police behavior.

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How a GAN Learns Realistic Mugshots

$$z \sim p_z, \quad \tilde{x} = G(z)$$

1. The **generator** G maps a random latent vector z into a synthetic mugshot $G(z)$.
2. The **discriminator** C sees real and generated mugshots and predicts real versus fake.
3. Training alternates: C improves detection, while G improves at fooling C .
4. After training, keep G . It learns $p(x)$ without detention labels.



Illustrative synthetic faces

The GAN learns what realistic mugshots look like. The separate model $m(x)$ predicts detention.

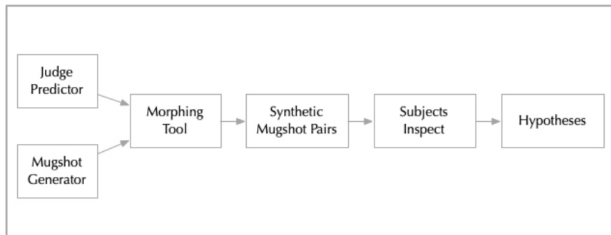
Why the Generator Matters

$$\underbrace{x \in \mathbb{R}^{786,432}}_{\text{all pixel arrays}} \quad \text{versus} \quad \underbrace{x = G(z), z \in \mathbb{R}^{512}}_{\text{learned face manifold}}.$$

- Almost every point in pixel space is noise, not a realistic face.
- A pixel gradient $\nabla_x m(x)$ identifies sensitive pixels, but following it can leave the face manifold.
- The paper instead changes z using $\nabla_z m(G(z))$. Every new image is generated as $G(z)$.
- The predictor m chooses the detention-related direction. The generator G keeps the images realistic.

Face manifold: the thin, learned subset of pixel space containing realistic mugshots.

Morphing and Interpretation



Starting from a common seed z_0 :

$$z^{\pm} \leftarrow z^{\pm} \pm \alpha \nabla_z m(G(z^{\pm})).$$

Stop when

$$m(G(z^-)) \leq 0.10,$$

$$m(G(z^+)) \geq 0.35.$$

1. Keep the generator and predictor fixed.
2. Show many pairs to independent subjects.
3. Ask subjects to name the contrast.

Validation Exercise: Age

Before interpreting detention, the authors test the communication protocol on a known target.

- Train an age predictor and morph synthetic face pairs along its gradient.
- **Task:** “Which face shows the hidden characteristic more strongly?” Age is never named.
- After each choice, give correct/incorrect feedback. Treat the first 10 pairs as learning trials.
- **Final 40 pairs:** 97.8% choose the face the model ranks as older.
- After all 50 pairs, ask which facial features drove the algorithm’s choices. 81.5% mention age or hair color.

By the evaluation rounds, subjects can recover a known feature from the morphs.

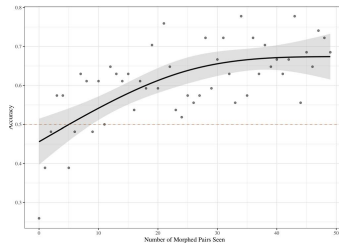
People Learn the Algorithm's Visual Signal



(a) Side-by-side mugshot morphs with detention probabilities of 0.41 and 0.13 respectively



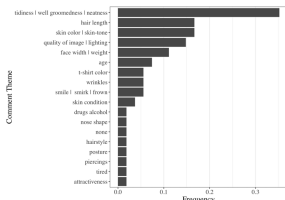
One seed, different predicted detention risk



Accuracy across successive pairs

- **Task:** “Which synthetic defendant is more likely to be detained?” The outcome is named.
- 54 subjects see 50 pairs and receive correct/incorrect feedback after every choice.
- Treat the first 10 as learning examples. Analyze accuracy on the final 40.
- Accuracy begins near chance, rises by more than 10 pp after 20 pairs, and reaches about 67% after 30.

First Discovery: Well-Groomed

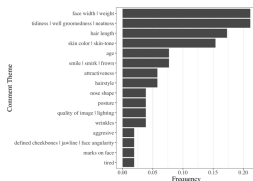


- **Discover:** After all 50 pairs, nearly 40% describe grooming-related differences.
- **Measure:** Comments only name the feature. 343 new workers rate 32,881 actual mugshots, producing $h_1(x)$.
- **Validate:** A one-SD increase in $h_1(x)$ predicts 1.74 pp lower model-predicted detention.
- **Explanatory power:** $R^2 = 0.0247$, or 11% of the full model's $R^2 = 0.2361$.

Second Discovery: Heavy-Faced



(a) A word cloud of the comments



- **Remove the prior signal:** Match detained and released training mugshots on grooming and skin tone. Refit m and generate “orthogonalized” morphs.
- **Discover:** 52 new subjects inspect 50 pairs. About 22% describe a wide, round, full, or heavy face. Call this $h_2(x)$.
- **Measure:** 343 independent workers rate actual validation mugshots on this feature.
- **Validate:** A one-SD increase in $h_2(x)$ predicts 2.17 pp lower model-predicted detention. $R^2 = 0.0384$: $h_2(x)$ explains 3.84% of variation in the refitted model’s scores.

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The New Features Predict Judge Decisions

Feature	Effect of 1 SD	Share of base rate	Adjusted R^2
Well-groomed	−2.0 pp	−8.9%	0.0021
Heavy-faced	−2.8 pp	−12.0%	0.0042
Both together			0.0061

- The corresponding quartile gaps are 5.5 pp for well-groomed and 7.0 pp for heavy-faced, versus 4.8 pp for violent charges.
- Both novel features explain 28% as much variation as the full set of observed variables excluding the image algorithm.
- Heavy-faced remains significant after controlling directly for the algorithm's prediction.

Regression Table

Practitioners Do Not Appear to Know the Features

- Fifteen public-defender and legal-aid staff provide 360 forecasts from pairs of actual mugshots.
- Their forecasts are much more accurate than ordinary subjects' forecasts.
- Their open-ended explanations rarely mention well-groomed or heavy-faced.
- Practitioner forecasts and the algorithm contribute nearly additive R^2 :

$$0.0165 \text{ (practitioner)} + 0.0166 \text{ (algorithm)} \approx 0.0338 \text{ (both)}.$$

Experts and the algorithm appear to use different signals.

What the Paper Does and Does Not Establish

Established in this setting

- faces predict judge decisions
- the algorithm contains novel signal
- morphs communicate part of that signal
- discovered features predict decisions

Left open

- field causal effect on judges
- exact psychological mechanism
- transport across jurisdictions
- completeness of the discovered feature set

The method ranks ideas for causal testing. It does not replace identification.

Assessment

- **Contribution:** a concrete protocol for turning predictive signal into interpretable candidate hypotheses.
- **Technical risk:** the generator limits what can be discovered, while human naming may compress correlated changes into one label.
- **Empirical risk:** single-county correlations and a lab experiment with non-judges do not establish judge-level causality.
- **Scope:** the method travels when outcomes are predictable, inputs contain interpretable signal, and a generator can create realistic nearby counterfactuals.

Conclusion

- Machine learning can contribute to science before formal hypothesis testing begins.
- Prediction supplies empirical discipline; human interpretation supplies portable language.
- Generative counterfactuals reduce irrelevant variation and reveal contrasts people can name.
- In the application, this process uncovers two facial features associated with detention decisions.

Algorithms find patterns. People name hypotheses. Research designs test them.

Appendix

Proposition 1: Conditions

Let P_c denote human discovery from n raw matched pairs and P_m discovery from n morph pairs.

The result requires:

1. people inspect no more than N pairs;
2. the share of interpretable hypotheses that are true is sufficiently small;
3. an implausible hypothesis fits a random morph pair with sufficiently low probability;
4. the human procedure's prior favors true hypotheses relative to random choice.

Then morphing weakly improves plausibility and replicability while preserving comprehensibility.

The proposition is an existence result. Its assumptions describe when nearest counterfactuals remove enough irrelevant variation to beat raw human comparison.

Discovery-Procedure Definitions

For a matched pair $e = ((y_0, x_0), (y_1, x_1))$ with $y_0 = 0$ and $y_1 = 1$, hypothesis h is consistent if

$$h(x_0) < h(x_1).$$

For a set of matched pairs E :

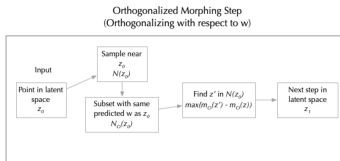
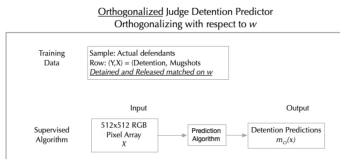
$$\mathcal{H}_E = \{h \in \mathcal{H} : h \text{ is consistent with every } e \in E\},$$

$$\mathcal{I}_E = \mathcal{I} \cap \mathcal{H}_E.$$

The human procedure samples n pairs and selects from $\mathcal{I}_E$, possibly using an advantageous prior over Φ^* .

The morphing procedure replaces raw outcome pairs with nearest generated pairs that differ in the out-of-sample predictor $m(x)$.

Orthogonalized Morphs



- Match detained and released observations on grooming and skin tone.
- Fit a new image predictor in the matched sample.
- Morph along its detention gradient.
- Conduct a local random search around the endpoint.
- Require predicted covariates to match within 0.001.



The orthogonalized predictor generates a 27% versus 7% predicted detention contrast while limiting differences in the first discovered feature and skin tone.

Selected Summary Statistics

Variable	Train + validation	Training	Validation
Judge detains defendant	0.234	0.234	0.233
Re-arrest before disposition	0.251	0.250	0.251
Male	0.787	0.789	0.782
Black	0.694	0.693	0.695
Age	31.793	31.859	31.631
Violent charge	0.343	0.343	0.343
Felony charge	0.424	0.422	0.428

Training and validation samples are balanced across the reported variables. A multivariate test of equality yields $p = 0.42$.

Algorithm Prediction and Known Features

Predictor or contrast	Estimated effect	Interpretation
Male	0.1140	11.4 pp higher predicted detention
Skin tone, dark to light	−0.0417	4.2 pp lower across full scale
Attractiveness, 1 SD	−0.0051	0.51 pp lower
Competence, 1 SD	−0.0083	0.83 pp lower
Trustworthiness, 1 SD	−0.0041	0.41 pp lower
Dominance, 1 SD	0.0035	0.35 pp higher
Human guess	0.0399	3.99 pp higher

The full set of known features yields $R^2 = 0.2228$ for the image algorithm's prediction.

Novel Features and Judge Decisions

Specification	Heavy-faced	Well-groomed	Adjusted R^2
Heavy-faced only	−0.0234***		0.0042
Well-groomed only		−0.0198***	0.0021
Both features	−0.0226***	−0.0178***	0.0061
All known controls	−0.0218***	−0.0100*	0.0218
Controls plus image prediction	−0.0111***	−0.0022	0.0387

Coefficients describe a one-unit change on the nine-point human-rating scales. Standard errors cluster by arrestee. Selected columns are reproduced from the paper.

Randomized Lab Experiment

- Two samples of 500 US Prolific subjects, one per facial feature.
- Subjects first complete 15 training choices using actual cases and feedback.
- The test phase shows synthetic pairs with randomized feature direction and assigned structured case information.
- Pair fixed effects compare two variants of the same underlying synthetic pair:

$$y_{is} = \gamma_0 + \gamma_1 v_{is} + \delta_s + \varepsilon_{is}.$$

- 18,269 choices in the grooming experiment and 18,548 in the heavy-faced experiment.
- Randomized appearance changes detention recommendations by 1.3 pp and 1.9 pp, respectively.

Ethical and Measurement Considerations

- The exercise predicts **judge behavior**, not defendant criminality.
- Re-arrest is an imperfect risk measure shaped by police decisions and selective observation.
- The paper displays synthetic images and avoids presenting actual defendants.
- GAN quality determines which visual dimensions the method can discover.
- Human labels can compress correlated appearance changes into a single verbal category.
- A discovery tool can expose bias, but it can also create harmful stereotypes if correlations are misread as traits or causal facts.